Chapter 1: Solutions

Formula Sheet + Tips & Tricks

Formula Sheet

I. Vapour Pressure – Raoult's Law

Raoult's Law (Relative Lowering of VP)

$$\frac{P_A^\circ - P_A}{P_A^\circ} = x_B$$

P_A° = vapour pressure of pure solvent; P_A = vapour pressure of solution; x_B = mole fraction of solute.

Raoult's Law (alternate forms)

$$\frac{P_A^\circ - P_A}{P_A^\circ} = \frac{n_B}{n_A + n_B} = \frac{w_2 M_1}{w_1 M_2}$$

n_A, n_B = moles of solvent, solute; w_1, w_2 = masses; M_1, M_2 = molar masses of solvent, solute.

Raoult's Law for Volatile Liquids

$$P_{\text{total}} = P_A^\circ x_A + P_B^\circ x_B$$

Total vapour pressure of an ideal solution of two volatile liquids; also $P_i \propto x_i$.

II. Colligative Properties

Elevation in Boiling Point

$$\Delta T_B = T_B - T_B^\circ = K_b \cdot m$$

K_b = molal elevation (ebullioscopic) constant; m = molality of solution.

Depression in Freezing Point

$$\Delta T_f = T_f^\circ - T_f = K_f \cdot m$$

K_f = molal depression (cryoscopic) constant; m = molality of solution.

Osmotic Pressure

$$\pi = MRT$$

M = molarity (mol/L); $R = 0.0821$ L atm mol^{-1}K^{-1}; T in kelvin. Also written $\pi = CRT$.

Molarity

$$\text{Molarity} = \frac{\text{moles of solute}}{\text{volume of solution (L)}}$$

Used directly in the osmotic pressure formula.

III. van't Hoff Factor (Abnormal Colligative Properties)

van't Hoff Factor – Modified Formulas

$$\Delta T_B = K_b\, m\, i \quad \Big| \quad \Delta T_f = K_f\, m\, i \quad \Big| \quad \pi = MRT\, i \quad \Big| \quad \frac{P_A^\circ - P_A}{P_A^\circ} = x_B\, i$$

Multiply the normal colligative property expression by i to account for dissociation/association.

van't Hoff Factor Definition

$$i = \frac{\text{Normal molar mass}}{\text{Observed molar mass}} = \frac{\text{Observed colligative property}}{\text{Normal colligative property}} = \frac{\text{Total moles after diss./assoc.}}{\text{Total moles before diss./assoc.}}$$

$i < 1$ indicates association; $i > 1$ indicates dissociation.

Degree of Dissociation

$$\alpha = \frac{i-1}{n-1}$$

n = number of ions/particles produced per formula unit on complete dissociation.

Degree of Association

$$\alpha = \frac{i-1}{\frac{1}{n}-1}$$

n = number of particles that associate into one aggregate.

Tips, Tricks & Hacks

- **Strong electrolytes** ($>50\%$ dissociation, ionic compounds) dissociate almost 100%; **weak electrolytes** exist as a mix of ions and molecules.
- Common i values: glucose/sucrose/urea = 1; NaCl, KCl, HCl, $MgSO_4$, $CaCl_2^*$ = 2; Na_2SO_4, K_2SO_4, $CaCl_2$ = 3; $AlCl_3$ = 4 (assuming 100% dissociation).
- Quick reference molar masses: H_2O = 18, glucose $C_6H_{12}O_6$ = 180, urea CH_4N_2O = 60, sucrose $C_{12}H_{22}O_{11}$ = 342, acetic acid = 60, ethanol = 46, benzene C_6H_6 = 78, naphthalene $C_{10}H_8$ = 128, camphor $C_{10}H_{16}O$= 152, NaCl = 58.5, KCl = 74.5, $CaCl_2$ = 111, $MgSO_4$ = 120, Na_2SO_4 = 142, $AlCl_3$ = 133.5.
- In osmosis, solvent flows from the region of *lower* concentration (higher water potential) to *higher* concentration until equilibrium.
- Remember the order for boiling point elevation and freezing point depression setups: always define solution vs. pure solvent temperature carefully – $\Delta T_B = T_B - T_B^{\circ}$ (solution minus solvent), but $\Delta T_f = T_f^{\circ} - T_f$ (solvent minus solution).